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Tinnitus: clinical measurement

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Clinician A: “Why should we spend any time measuring tinnitus? How can you measure something that’s completely subjective? And besides, what good does it do to measure it?”

Clinician B: “It’s true that subjective experiences are harder to measure, but there’s now a large literature on how to measure tinnitus. And it *is* important to measure it in order to document the patient’s status before any treatment is started—otherwise it’s very difficult to show that your treatment has really changed anything.”

This presentation discusses general diagnostic guidelines for use in evaluating clinically significant tinnitus. “Clinically significant” means tinnitus that is severe enough that the patient has sought some type of professional help.

Tinnitus is just one of many health conditions for which there is presently no cure, although there are treatments that can offer some degree of relief. In such cases, better information about the nature of the problem is essential. The better the information gained about any health problem, the more likely it is that more effective treatment will ultimately be developed [1]. It is for this reason that tinnitus measurement procedures have been emphasized by many investigators since the 1930s [2].

Treatment for tinnitus, just as for any medical condition, depends on a careful diagnostic evaluation of the patient’s status. The tinnitus tests included here should be viewed as providing the scaffolding on which a comprehensive approach to diagnosis and treatment can be organized. Each clinician will undoubtedly evolve his or her own refinements or additions to the basic set of measures described in this article.

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Measurement goals

The tests important for clinical use are designed to do the following:

1. Establish baseline characteristics of the tinnitus
2. Define the patient's treatment needs
3. Offer guidance in selecting specific treatments
4. Provide quantitative measures for monitoring treatment effects

These four clinical activities can also be described under the headings *intake measures*, *treatment goals*, *treatment strategies*, and *outcome measures*.

It should be noted that quantification and documentation are as essential for working with tinnitus as they are in audiometry and other standard forms of audiologic practice. Further, documentation of standardized tinnitus measures is the only way in which clinicians working in the area of tinnitus will be able to learn from each others' experience and, by sharing their own clinical insights, contribute to the benefit of all [3].

Development of the measures

The set of tinnitus measures presented here has been standardized over the past 20 years of use in the Oregon Tinnitus Clinic [4]. Before that, the CIBA Foundation symposium on tinnitus advocated these measures as a standard test battery, to have a basis for comparability between clinics [5]. Four of the measures involved audiometric equipment and consisted of the following: (1) tinnitus pitch, (2) tinnitus loudness matches using external tones, (3) maskability, and (4) residual inhibition (RI). Although the usual audiometer can perform much of the testing described below, tinnitus testing is enhanced by the addition of bench equipment or by use of specially designed tinnitus synthesizers.

For the past decade, testing in the Oregon Tinnitus Clinic has used a computer-based system known as Computer Auditory Research Laboratory [6]. Further information about a more recently developed system is also available.

The standardized test battery used at the Oregon Tinnitus Clinic has generated a wealth of information based on the Tinnitus Data Registry [7–10]. The Registry is a large computerized database set up in 1981 to provide statistical comparison and evaluation of data obtained from the patients attending the Tinnitus Clinic. The information available in statistical form has provided a number of insights linking tinnitus measures to the corresponding treatment decisions, and has provided the basis for the present discussion.

Tinnitus clinic tests

Audiometric

Before any tinnitus testing the patient's audiogram is determined for each ear at standard frequencies from 100 through 8000 Hz, and even higher if possible (the authors use a special high-frequency audiometer and test up to 16,000 Hz if the patient's hearing permits). The audiometric range is extended as far as possible for the simple reason that most tinnitus is high-pitched, as shown later.

Tinnitus localization

Before the tinnitus tests, the localization of the patient's tinnitus must be established [11]. The tinnitus localization affects the selection of which ear to test. If the tinnitus is unilateral or is bilateral but louder on one side, the authors prefer to test the contralateral ear because there is less likelihood of interference between the test stimuli and the tinnitus. If the tinnitus is bilateral and equal, selection of the test ear is based on other factors, such as the extent of hearing loss, recruitment, or diplacusis, or any other conditions that might affect pitch discrimination.

Tonal versus noise-like tinnitus

In general tinnitus can be divided into two broad categories: tonal tinnitus, often described as ringing or whistling; and noise-like tinnitus, often described as roaring or hissing. A few patients have combinations of these. Roaring tinnitus tends to be relatively low-pitched noise, whereas hissing is usually relatively high pitched. In a review of several thousand tinnitus patients seen at the authors' tinnitus clinic, three fourths described their tinnitus as being more like a tone than a noise. This information is important because the procedure for pitch and loudness matching differs slightly for tonal versus noise-like tinnitus.

Test 1: measuring the pitch of tinnitus

Tonal tinnitus

The authors identify the pitch of tinnitus using a two-alternative forced-choice method. In comparison 1, the examiner starts by comparing 1000 and 2000 Hz, with the loudness of each tone adjusted independently so as to match the loudness of the tinnitus. The two tones are presented alternately to the patient, several times if necessary, and the patient indicates which tone is more like the tinnitus. Care must be taken to ensure that the lower-frequency tone is not always presented first, but rather that the order of presenting the two tones in each pair is varied at random.

If in comparison 1 the patient selects 1000 Hz as the better match, it is likely that the tinnitus pitch is relatively low; one then compares different

frequencies at 100-Hz intervals in the vicinity of 1000 Hz to find the best match.

It is much more common for tinnitus to be high pitched, so patients are most likely to select 2000 Hz as the better match. In that case comparison 2 compares 2000 and 3000 Hz, again adjusting the new tone so that its loudness is matched to that of the tinnitus. Continuing in this manner, eventually a tone pair is presented in which the lower tone rather than the higher tone is judged the better match (Table 1).

For example, if 4000 Hz is selected as being a better match than 5000 Hz, one concludes that the tinnitus pitch is somewhere in the 4000-Hz region. Call this the “provisional pitch match” for the tinnitus because a further test is now required, the test for octave confusion, which is needed because octave confusions occur very commonly in all tests of frequency discrimination, even with normally hearing subjects who do not have tinnitus. In the present example, one compares 4000 and 8000 Hz. Octave confusion has been observed to occur about 60% of the time in the authors’ tinnitus clinic. In the testing example cited previously, there is a good chance that the patient would select 8000 Hz as the best match for the tinnitus.

It is clear that there are some situations in which the tone selected by the patient is too high to perform an octave confusion test. This situation arises if the comparison procedure yields a provisional pitch match that is high enough (eg, 5000 Hz) that the examiner does not have the capability of generating tones an octave higher (at 10,000 Hz in this example). The problem also arises if the patient’s high-frequency hearing is so poor that, even if one can generate a 10,000-Hz tone, it is inaudible to the patient. If a high-frequency audiometer or other high-frequency tone generator is available, however, it is always desirable to test for octave confusion even if there is doubt that the patient will be able hear the highest tones. Tonal tinnitus is seldom very low pitched; it is much more commonly matched to a relatively high-pitched tone.

Table 1
Sample data sheet using the two-alternative forced-choice procedure for matching the pitch of tinnitus

Comparison number	Test tones, Hz	Better-matching tone
1	1000 vs 2000	2000
2	2000 vs 3000	3000
3	3000 vs 4000	4000
4	4000 vs 5000	5000
5	5000 vs 6000	5000*
6	5000 vs 10,000†	10,000
Best match for tinnitus pitch:		10,000 Hz

* Comparison number 5 identifies the provisional pitch match (5000 Hz).

† Comparison number 6 is the octave confusion test, which succeeds in identifying the best pitch match at one octave above the provisional pitch-matching frequency.

Noise-like tinnitus

The procedure for matching the pitch of noise-like tinnitus is essentially the same as for tones, except that the matching sounds are not tones but, rather, narrow bands of noise (eg, one-third octave bands) centered at the frequencies previously indicated.

Many clinics do not have equipment capable of generating frequency-selectable, narrowband noise. If such equipment is not available, then a practical compromise is to ask patients with noise-like tinnitus to match their tinnitus to comparison tones in the same manner as is done for tonal tinnitus. This matching task is obviously a more difficult one for patients than if comparison noisebands can be provided; however, patients often comment that, as the frequency of noisebands rises, they sound too “wide” and should be made narrower in order to match the tinnitus. This suggests that at high frequencies, particularly where there is significant hearing loss, patients may have difficulty differentiating between tones and noise.

Pitch-matching data from the tinnitus clinic

The authors previously reported pitch-matching data obtained from a large series of Tinnitus Clinic patients as summarized in Table 2 [7]. It can be seen that 75% of the patients matched their tinnitus pitch at or above 4000 Hz. The median pitch for the entire group of 1519 patients was 6000 Hz.

An interesting case occurs with tinnitus described as crickets. Such tinnitus can usually be matched with a combination of two relatively high-frequency tones presented simultaneously, such as 4000 plus 4060 Hz or 3000 plus 3040 Hz .

Potential treatment effects on tinnitus pitch

It is not uncommon for tinnitus treatments to reduce the perceived pitch of the tinnitus [12], even if the loudness is not greatly affected. Such pitch reductions render the tinnitus generally more tolerable. This beneficial effect should be documented as a desirable treatment outcome. For example,

Table 2
Pitch matches obtained using a tinnitus synthesizer (range 100–16,000 Hz)

Frequency, Hz	Number of patients	Percent
≤1900	139	8.9
2000–3900	244	16.1
4000–5900	302	19.9
6000–7900	285	19.8
8000–9900	345	22.8
10,000–16,000	204	13.5
Total	1519	100

Data were obtained from an overall series of 1630 patients who attended the Tinnitus Clinic during 1986–1995, of whom 1514 were able to provide pitch matches for their tinnitus.

patients may react positively to reduction of the tinnitus pitch by saying such things as “even though the tinnitus is still there, the masker ‘takes the edge off’ the tinnitus and it is no longer so disruptive.”

Another potentially desirable treatment effect may occur in patients with multiple tinnitus sounds that differ in regard to pitch. The treatment may eliminate some of the sounds, usually the highest pitched, while leaving lower-pitched tinnitus relatively unchanged. Again, patients are likely to regard this as a positive treatment effect, and it should be so documented.

Test 2: loudness matching for tinnitus

A number of loudness matches are obtained at the different matching-tone frequencies ranging from 1000 Hz to the frequency matching the pitch of the tinnitus. In the authors’ clinic, this series of loudness matches at different frequencies is recorded in dB Sensation Level (SL, dB above threshold). It can, for example, be recorded by the examiner on a standard audiogram form, although the frequency range may need to be extended above 8000 Hz.

It has become conventional to report tinnitus loudness matches at one or both of the following: the frequency matching the pitch of tinnitus and at 1000 Hz. Table 3 illustrates the range of loudness matches at the tinnitus pitch.

The rationale for reporting loudness matches at 1000 Hz is that, in many patients, the loudness matches at the tinnitus pitch are in general quite small, whereas those at 1000 Hz are substantially larger. The restriction of range at the tinnitus pitch occurs because the loudness matches tend to decrease as the comparison tones approach the frequency matching the tinnitus pitch; in a sense, they converge toward the tinnitus frequency. As Table 3 shows when using external tones matched to the tinnitus pitch, almost 70% of the loudness matches are at or below 6 dB SL, and 84% are below 9 dB SL. These are such small decibel values that many have

Table 3
Tinnitus loudness matches using a loudness-balance procedure to an external tone matching the pitch of the tinnitus

Loudness match, dB SL	Number of patients	Percent
0–3	572	40.2
4–6	408	28.7
7–9	212	14.9
10–12	99	7
13–15	64	4.5
16–18	30	2.1
≥19	37	2.6
Total	1422	100

Data were obtained from the same series of 1630 patients shown in Table 1; 1422 of these patients were able to provide loudness matches for their tinnitus.

wondered how tinnitus patients can complain of very loud tinnitus when their loudness matches are so low.

The phenomenon of loudness recruitment can partly explain this paradox, as first suggested by Fowler [13,14] and later by Vernon [15]. Goodwin and Johnson [16] then showed that loudness matches at 1 kHz (where hearing in their subjects was normal) were in fact larger. Data from the Tinnitus Clinic have also confirmed that result. Even when recruitment is taken into account, however, the loudness matches at 1 kHz are still considerably smaller than predicted based on the strong negative reactions they evoke from patients [17]. Evidently the paradox will only be resolved by further study of the neurologic processes underlying loudness perception in individuals with tinnitus. It is perhaps not surprising that there are discrepancies between measurements of real physical sounds versus measurements of the phantom sounds referred to as tinnitus.

Use of loudness matches to confirm the presence of tinnitus

In situations where compensation is sought for tinnitus, for example following occupational hearing damage or automobile injuries, it is necessary to evaluate the validity of the complainant's claim. Because individuals who truly have tinnitus are usually quite reliable in their ability to match its loudness (because they are listening to an internal standard), in litigation cases one should determine the loudness match of the claimed tinnitus five or six times and record the loudness matches so obtained [18]. The various loudness tests should be separated by small intervals in which other tests can be done as long as they do not alter the individual's tinnitus. If this set of loudness matches does not vary by more than about 3 dB, that has been accepted as court's evidence that the patient probably has tinnitus as claimed.

Potential treatment effects on tinnitus loudness matches

It has been shown that, when patients reported beneficial effects of tinnitus treatment using the benzodiazepine alprazolam, their tinnitus loudness matches showed corresponding reduction [19]. Similar effects have been observed with tinnitus masking [12,20]. These results are not surprising, even though it is well known that there is no correlation between loudness matches and the perceived loudness of tinnitus [13–15,21]. The reason for the lack of correlation is that the loudness matches exhibit a severe restriction of range, which means that it is difficult to demonstrate any correlation because the range of possible reductions is very small in the case of the loudness matches. Nevertheless, it is possible for even small values to show consistent reductions following treatment. As a case in point, in the alprazolam study cited previously, the loudness matches in the drug-treated group decreased significantly, from a pretreatment mean of 7.5 dB to

a posttreatment mean of 2.3 dB. In the placebo-treated group, the post-treatment loudness matches were essentially unchanged.

Test 3: determining the minimum masking level

Masking of tinnitus by external sounds has been described by a number of investigators, starting in the 1800s. The modern era of masking studies began with the work of Feldmann [22,23]. His observations indicated that approximately 35% of tinnitus patients were best masked by external sounds at the tinnitus frequency. Subsequently, Mitchell [24] found that about 50% of patients that he tested showed the same type of frequency dependence. From these studies one can conclude that tests for maskability should be done using stimuli that include energy at frequencies matching patients' tinnitus. Because most tinnitus is high pitched it is important that the noise used to determine maskability include the high frequencies. In the authors' clinic the stimuli used to determine the minimum masking level (MML) is a band of noise from 2000 Hz extending through 12,000 Hz.

The masking noise is presented to the affected ears under earphones, starting at zero amplitude; the amplitude is then slowly increased while the patient is asked to indicate when the noise first becomes audible (determining threshold for the noiseband). The sound amplitude is then increased in small steps (preferably 1 to 2 dB, up to the maximum tolerable level for the patient) until the patient reports the tinnitus is no longer audible in that ear. That effect is referred to as *complete masking* and the corresponding sound level is defined as the MML and is recorded in dB SL [25]. In about 5% of patients, the tinnitus may not be covered completely but instead its loudness may be reduced by masking; that effect is referred to as *partial masking*, and in another 5% there is no diminution of tinnitus during the masking test [25]. In patients with bilateral tinnitus the procedures described previously are attempted for each ear independently, and also, if feasible, when both ears are stimulated simultaneously. In all cases, MMLs are reported in dB SL.

Table 4 shows the MMLs measured in 1383 tinnitus patients. It is clear from these data that in most (57.7%), masking occurs at or below 9 dB SL, which is a small figure. Not only is 9 dB SL a relatively soft sound but its frequency composition (2 to 12 kHz) is such that it does not interfere with speech comprehension. Similar statements apply to wearable masking devices; although they do not reach as high as 12 kHz, they are generally higher than most speech frequencies and they do not generally interfere with the wearer's speech comprehension.

Effects of treatment on MMLs

Minimum masking levels have been shown to be reduced in patients who experienced reductions in the severity of their tinnitus [12]. Such data help to reinforce the conclusion that the treatment was beneficial. It is also likely

Table 4

Minimum masking levels for a large sample of Tinnitus Clinic patients

MML, dB SL	Number of patients	Percent
0–3	270	19.5
4–6	309	22.3
7–9	220	15.9
10–12	175	12.7
13–15	104	7.5
16–18	74	5.4
19–21	54	3.9
22–24	44	3.2
25–27	25	1.8
28–30	25	1.8
>30	83	6
Total	1383	100

Data were obtained from the same series of 1630 patients shown in Table 1; a total of 1383 of these patients were able to identify minimum masking levels for their tinnitus.

that patients with treatment-reduced MMLs more easily experience masking induced by exposure to ambient environmental noise, and that fact in itself should provide additional relief. Any procedure that tends to reduce the patient's MMLs should be viewed as having merit. Likewise, documentation of changes in MMLs can provide important information to those wishing to compare the effectiveness of different treatments, or of a treatment versus a placebo.

Test 4: measurement of residual inhibition following masking

Residual inhibition (RI) is the reduction of tinnitus after a period of masking. It is important to inform the patient that RI is not a form of therapy, but rather it is simply additional information about their tinnitus. In the Oregon Tinnitus Clinic, RI is induced by applying a masking noise at MML plus 10 dB for 60 seconds, presented under earphones. At the end of the 60 seconds the masking sound is removed and the patient is asked to indicate how their tinnitus sounds. If the tinnitus seems to be altered following the offset of masking, the patient is then asked to report when the tinnitus begins to return to its normal level, and when it has completed the return to the normal level. *Complete RI* is the term used to describe the complete absence of tinnitus following the offset of masking, which occurs in some patients, and *partial RI* describes the period during which the tinnitus is present at a reduced loudness level and is in the process of returning to its normal level [25]. The clinician times the duration of these recovery periods using a stopwatch. Table 5 shows the relative occurrence of these recovery processes in the authors' Tinnitus Clinic patients.

From these data it is clear that most (88%) patients display some form of RI. Patients often respond to the RI test with great surprise, and even with

Table 5
Reported extent of residual inhibition following 1 minute of broadband noise masking

RI type	Number of patients	Percent
No RI of any type	173	11.9
Partial RI only	476	32.8
Complete RI only	34	2.3
Complete RI + Partial RI	768	52.9
Total	1451	100

RI, residual inhibition.

Data were obtained from the same series of 1630 patients shown in Table 1; 1451 of these patients were able to participate in a test for RI in at least one ear.

emotional reactions, such as tears. For many patients this is the first time they have experienced silence for many years. It is a very dramatic and valuable demonstration to them that their tinnitus is not immutable.

From Table 5 it can be seen that the most unusual situation is that of complete RI occurring alone, without being followed by a period of partial RI. These patients have a complete absence of tinnitus but when it returns it does so in one sudden and complete stroke.

The duration of RI for each patient who experiences it is calculated as the sum of the times for complete RI and partial RI for that patient. From Table 6 it can be seen that RI durations vary widely, with most being of relatively short duration but with a small number of patients who experience longer durations. Occasionally, the duration of RI exceeds an hour or even longer, and in such cases it may be necessary to terminate further testing because the tinnitus is no longer there to be tested! It is always surprising that a 1-minute period of masking is capable of producing such long-lasting effects. The RI phenomenon is completely unexplained, insofar as the underlying neurologic processes are concerned, and it remains one of the most interesting and provocative challenges facing tinnitus researchers.

Doubling the duration of masking does not seem to cause systematic changes in the RI durations [18]. In the authors' experience, alterations of

Table 6
Measured duration of residual inhibition following 1 minute of broadband noise masking

RI duration (minutes)	Number of patients	Percent
0.1–1.9	554	56.8
2–3.9	224	23
4–5.9	109	11.2
6–7.9	40	4.1
8–9.9	24	2.5
>10	25	2.6
Total	976	100

RI, residual inhibition.

Representative results based on right ear data only.

the masking levels used to induce RI (eg, to MML plus 5, 15, or 20 dB) result in slightly less RI than does MML plus 10 dB [18]. To date it is not known how to extend the period of RI, although one would obviously like to be able to do that, and many tinnitus patients would benefit if such a technique could be developed. There have been some anecdotal reports that using tonal masking at the same pitch as the tinnitus, instead of using a masking noiseband, results in longer periods of RI; however, such reports await experimental confirmation.

The entire area of RI needs close scrutiny and investigation, because if it is possible to induce RI lasting a matter of a few days it would present a credible and highly desirable relief procedure [26]. With the present generation of wearable masking devices, even when they are optimally fitted, only about 75% of patients display RI when they remove the wearable devices. The reduced effectiveness of wearable maskers for producing RI, as opposed to the effectiveness of masking by bench equipment such as the tinnitus synthesizer used in the clinic, is also unexplained but may be caused by the differences between the frequency compositions of the noise in wearable versus bench equipment.

Treatment Effects on Residual Inhibition

To date, we are not aware of any treatment trials in which well-documented changes in the occurrence of RI have been reported. However there is substantial anecdotal evidence that longterm use of wearable masking devices can produce an “accumulation” of RI effects, such that patients note they experience longer and longer periods of RI after they take their wearable devices off. This informal information is sorely in need of systematic investigation to see how well it holds up under careful followup evaluation. There are certainly patients who report that they no longer need to wear maskers. It is possible, however, that in such cases the tinnitus would have diminished over time even without the use of tinnitus masking. Further study is clearly indicated.

NOTES

Note 1. Bench equipment for testing of tinnitus is readily assembled, consisting of: headphones with excellent high frequency response; tone and noise generators with variable frequency controls; precision dB attenuators; electronic switches to eliminate onset and offset transients; and low-noise amplifiers with appropriate frequency response (50 Hz to 20 kHz). It is desirable to have a setup that can provide two tones or two noisebands at once, each with its own independent switches, attenuation, and volume controls.

Note 2. Information on an improved and updated version of the Computerized Auditory Research Laboratory (CARL) may be obtained

from Dr. James Henry of the VA National Center for Rehabilitative Auditory Research, Portland Veterans Affairs Medical Center, Portland, OR (503-220-8262 ext 57466); email: Henry.James@med.va.gov

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